```
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;

ENTITY fsm IS
PORT (
      x, time1, time2, clk, rst: IN std_logic;
      y: OUT std_logic);
END ENTITY;

ARCHITECTURE fsm OF fsm IS
      TYPE state IS (A, B, C, D);
      SIGNAL present_state, next_state: state;
BEGIN
      PROCESS(clk)
      BEGIN
            IF(rising_edge(clk)) THEN
                  IF(rst = '1') THEN
                        present_state <= A;
                  ELSE
                        present_state <= next_state;
                  END IF;
            END IF;
      END PROCESS;

      PROCESS(x, time1, time2, present_state)
      BEGIN
            CASE present_state IS
                  WHEN A =>
                        IF x = '1' THEN
                              next_state <= B;
                        ELSE
                              next_state <= A;
                        END IF;
                  WHEN B =>
                        IF time1 = '1' THEN
                              next_state <= C;
                        ELSE
                              next_state <= B;
                        END IF;
                  WHEN C =>
                        IF x = '0' THEN
                              next_state <= D;
                        ELSE
                              next_state <= C;
                        END IF;
                  WHEN D =>
                        IF time2 = '1' THEN
                              next_state <= A;
                        ELSE
                              next_state <= D;
                        END IF;
            END CASE;
      END PROCESS;

            y <= '0' WHEN present_state = A or present_state = B ELSE
                  '1';
END ARCHITECTURE;
```

Q5- Consider the state diagram in Figure Q5

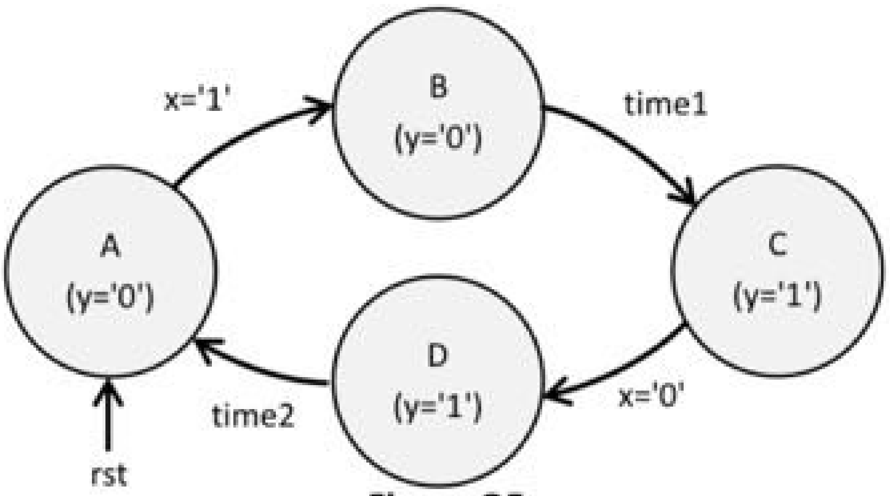
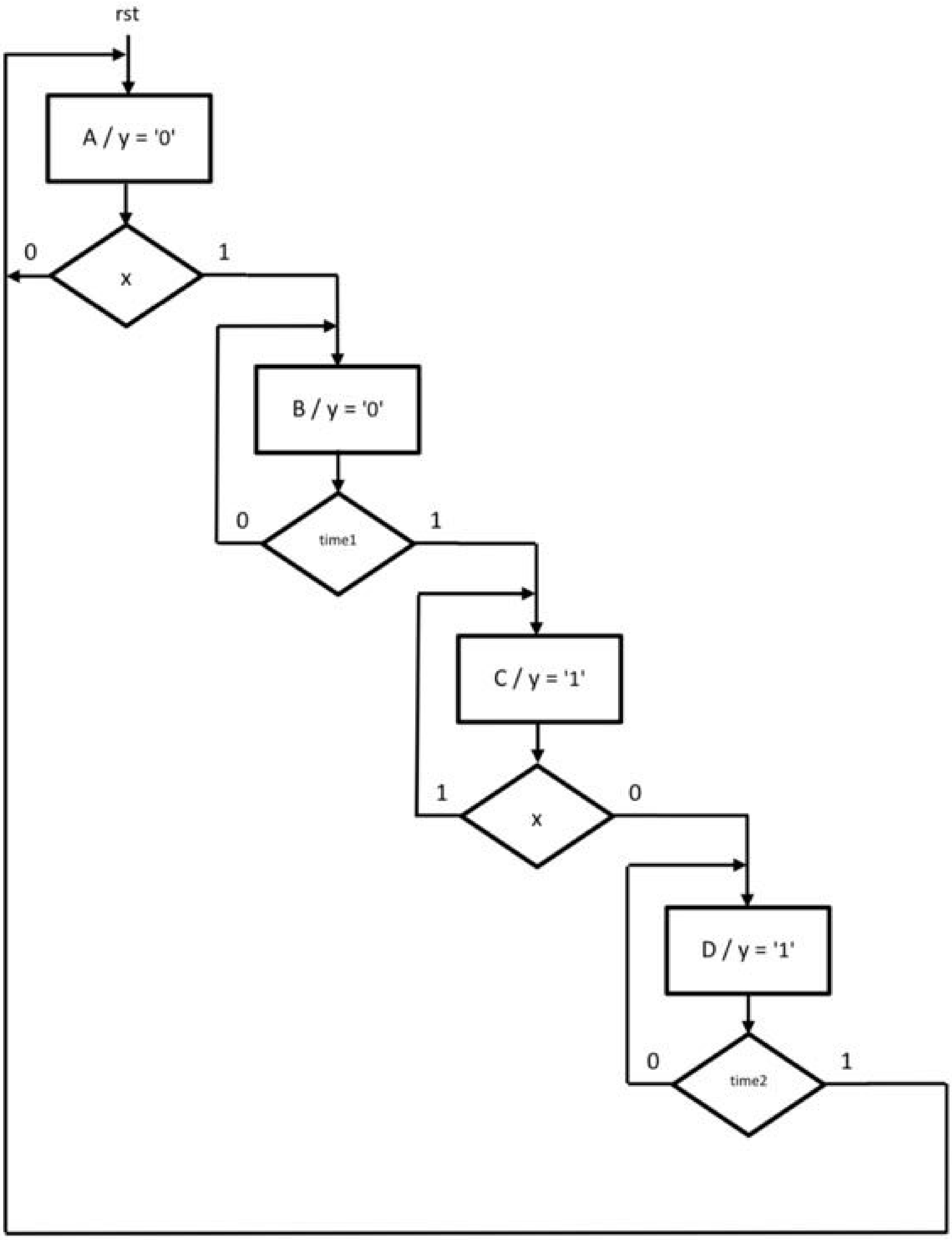


Figure Q5

(A) Draw the algorithmic state machine (ASM) of this state machine. (5 Marks)

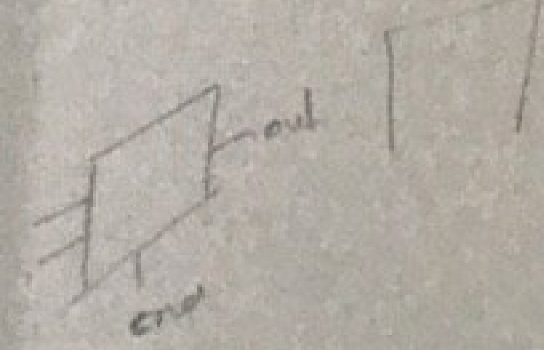


Ministry of Higher Education & Scientific Research		Subject: Introduction to digital lab
University of Kufa / College of Engineering		Year: Third
Department of Electronic and Communications Engineering		Time: Three hours
		Date: 12 / 11 /2019
		Examiner: Dr. Ammar Jallawi

Calculators are not allowed *Note: Answer all questions*

Q1: Say that a(7:0) = "00110011" and b(3:0) = "1111". Determine the values produced by the assignments below. (answer only 5) (5 marks)

a) a(7 DOWNTO 4) NAND "0111"
b) a(7 DOWNTO 4) XOR NOT b
c) "1111" NOR b
d) b(2 DOWNTO 0) XNOR "101"
e) a(2 DOWNTO 0) & "000"
f) a(5) & b(5) & a(5 DOWNTO 2)

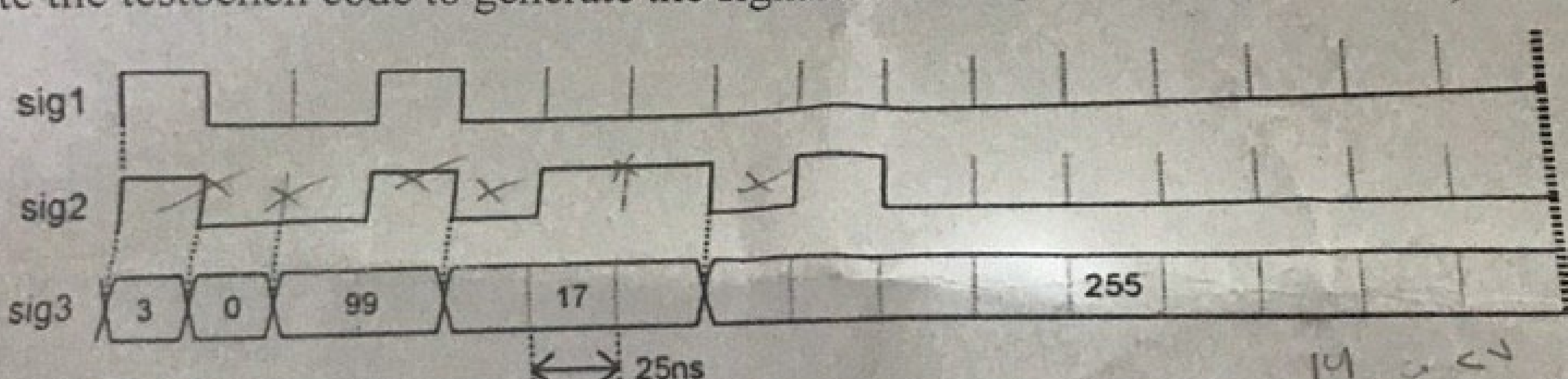


5432 1100

Q2: Implement a 4x1 multiplexer with: (10 marks)

A- where inputs are single bit and using only logical operators (gate level design).
B- N-bit inputs specify N using GENERIC and using WHEN statement.

Q3: Write the testbench code to generate the signals in the figure bellow: (10 marks)



Q4: Write the code for a zero to nine counter with reset signal input and the output (state value) is displayed using 7-segment displays HEX0. (15 mark

Dr. Ammar Jallawi Mahmood

Good luck although you shouldn't need it!

سؤال 4

Marked out of 5.00

Use the following code as a component in your top level design to build a **3-bit Subtractor** and compile it in Quartus II and upload a print screen showing the successful compilation with the code. **The top level :design entity name should be your student ID**

entity **FA** is

;port (A,B,Cin: in std_logic

;(S, Cout: out std_logic

;end **FA**

Architecture arch of **FA** is

begin

Cout <= Cout <= ((A xor B) and Cin) or (A ;(and B

;S <= (A xor B) xor Cin

;end arch

سؤال 5

Incorrect

Mark 0.00 out of 1.00

To design a 0 to 8 counter we will need how many D- ?flip flops

:Select one

مراجعة العودة

THEN
;slow_count <= slow_count + 1
;END IF
;END PROCESS

four-bit counter that uses a slow enable for selecting --
digit
(PROCESS (CLOCK_50
BEGIN
IF ( )

THEN
IF (slow_count = 0) THEN
IF (digit_flipper =) THEN

;"digit_flipper <= "0000
ELSE
;digit_flipper <= digit_flipper + 1
;END IF
;END IF
;END IF
;END PROCESS

;bcd <= digit_flipper
drive the display through a 7-seg decoder --
;(digit_0: bcd7seg PORT MAP (bcd, HEX0
;END Behavior

;LIBRARY ieee
;USE ieee.std_logic_1164.all

ENTITY bcd7seg IS
;(PORT (bcd : IN STD_LOGIC_VECTOR(3 DOWNTO 0
;((display : OUT STD_LOGIC_VECTOR(0 TO 6

Incorrect
Mark 0.00 out of 1.00

سؤال 5

To design a 0 to 8 counter we will need how many D-flip flops?

Select one:

a. 3

b. 5

c. 4

d. 6

Incorrect
Mark 0.00 out of 1.00

سؤال 6

The following VHDL code is for:

```
entity XXX is
port (B, D, R : in STD_LOGIC;
Q: out STD_LOGIC);
end XXX;
architecture arch of XXX is
begin
X: process (B, R)
begin
```

```
;bcd <= digit_flipper
drive the display through a 7-seg decoder --
;(digit_0: bcd7seg PORT MAP (bcd, HEX0
;END Behavior

;LIBRARY ieee
;USE ieee.std_logic_1164.all

ENTITY bcd7seg IS
;(PORT ( bcd : IN STD_LOGIC_VECTOR(3 DOWNTO 0
;((display : OUT STD_LOGIC_VECTOR(0 TO 6
;END bcd7seg

ARCHITECTURE Behavior OF bcd7seg IS
BEGIN

(PROCESS (bcd
BEGIN
CASE bcd IS
;"WHEN "0000" => display <= "0000001
;"WHEN "0001" => display <= "1001111
;"WHEN "0010" => display <= "0010010
;"WHEN "0011" => display <= "0000110
;"  " => WHEN "0100" => display

;"WHEN "0101" => display <= "0100100
;"WHEN "0110" => display <= "0100000
;"WHEN "0111" => display <= "0001111
;"WHEN "1000" => display <= "0000000
;"WHEN "1001" => display <= "0000100
;"WHEN OTHERS => display <= "1111111
;END CASE
;END PROCESS
;END Behavior
```

;Q <= D

;end if

;end if

;end process X

;end arch

:Select one

a. D latch with enable

b. D flip Flop with Asynchronous clear

c. D flip Flop with Asynchronous clear and enable

d. D- Flip flop without clear, reset and enable

e. D flip Flop with Synchronous reset and enable

f. D flip Flop with Synchronous reset

Incorrect
Mark 0.00 out of 1.00

سؤال 7

Given the signals bellow

;(signal X : std_logic_vector(7 downto 0

; D<= 0
;WAIT Clock_Cycle

;WAIT
;END PROCESS init

Clock_1 : PROCESS

BEGIN

Clock_Loop: Loop
;'Clk<= '0
;wait Clock_Cycle/2
;'Clk<= '1
;wait Clock_Cycle/2
end loop

;END PROCESS
;END part4_arch
:Select one

a. 1011 0010

b. 1110 0001

c. 0000 0000

d. 1110 0000

e. 1011 0111

Given the signals bellow

;(signal X : std_logic_vector(7 downto 0

;(signal Y :unsigned(7 downto 0

Which of the following is a legal VHDL assignment

:Select one

;(a. X <=unsigned(Y

;(b. Y <=to_unsigned(201, 8

;(c. Y <=std_logic_vector(X

;(d. X <=to_integer(Y

سؤال 8
Incorrect
Mark 0.00 out of 1.00

Say that a(7:0) "00110011" and b(3:0) = "1111". Determine the values produced by the :assignments below

;a(7 DOWNTO 4) XOR NOT b

:Answer

0011 xor not 1111 (0011) xor not (1111)

سؤال 9
Not answered
Marked out of 2.00

سؤال 1

Incorrect
Mark 0.00 out of 1.00

The truth table below represents a FSM with a 1-bit input (x) and a 1-bit output y. Suppose the machine is initialized to its starting state '00' and then processes the input values: 1,0,0,1,0,0 (1 input bit after each Clk). What are the final state?

Present State		Next State				Output	
		$x = 0$		$x = 1$		$x = 0$	$x = 1$
A	B	A	B	A	B	y	y
0	0	0	0	0	1	0	0
0	1	0	0	1	1	1	0
1	0	0	0	1	0	1	0
1	1	0	0	1	0	1	0

:Select one

a. 10

b. 01

c. 11

d. 00

سؤال 2

Incorrect
Mark 0.00 out of 2.00

Mark 0.00 out of 1.00

:The following VHDL code is for

```
entity XXX is
;port (B, D, R : in STD_LOGIC
;(Q: out STD_LOGIC
;end XXX
architecture arch of XXX is
begin
(X: process (B, R
begin
if (R='1') then
;'Q <= '0
else
if (rising_edge(B)) then
;Q <= D
;end if
;end if
;end process X
;end arch
```

:Select one

a. D latch with enable

```
;(
;END COMPONENT

BEGIN

i1 : 4SRegister
) PORT MAP
list connections between master ports and --
signals
,Clk => Clk
,D => D
,SCLR => SCLR
O => O
;(

init : PROCESS
BEGIN
;'SCLR<='0
;'D<='0
;WAIT 10ns
;'D<='1
;WAIT 15ns
;'D<='0
;WAIT Clock_Cycle
;'D<='1
;WAIT 2*Clock_Cycle
;'D<='0
;'SCLR<='1
;WAIT Clock_Cycle
;'D<='1
;WAIT 3*Clock_Cycle
```

Incorrect
Mark 0.00 out of 2.00

سؤال 2

Given the Testbench code bellow that aim to test a 8-bit shift register with serial input D, Clear input (SCLR operate on active high), and a clock (CLK). This shift register operate on +ev edge. Then what are the content of this register would be after 8 clock cycles :with shift left

```
;LIBRARY ieee
;USE ieee.std_logic_1164.all

ENTITY part4_vhd_tst IS
;END part4_vhd_tst
ARCHITECTURE part4_arch OF part4_vhd_tst IS
;Clock_Cycle: time := 40ns
;SIGNAL Clk : STD_LOGIC

;SIGNAL SCLR : STD_LOGIC
;SIGNAL D : STD_LOGIC

SIGNAL O : STD_LOGIC


COMPONENT 4SRegister

) PORT
;Clk D, SCLR: IN STD_LOGIC
O : OUT STD_LOGIC
;(

;END COMPONENT


BEGIN
```

**Use a 1-digit BCD counter enabled at 1Hz known --
that the clock frequency (CLOCK_50) is 25 MHz and
?all counters work on rising edge**

```
;LIBRARY ieee
;USE ieee.std_logic_1164.all
;USE ieee.std_logic_unsigned.all

ENTITY part3 IS
;PORT ( CLOCK_50 : IN STD_LOGIC
;((HEX0 : OUT STD_LOGIC_VECTOR(0 TO 6
;END part3

ARCHITECTURE Behavior OF part3 IS
COMPONENT bcd7seg
;(PORT ( bcd : IN STD_LOGIC_VECTOR(3 DOWNTO 0
;((display : OUT STD_LOGIC_VECTOR(0 TO 6
;END COMPONENT
;(SIGNAL bcd : STD_LOGIC_VECTOR(3 DOWNTO 0
SIGNAL slow_count : STD_LOGIC_VECTOR( [ ▼ ]

;(DOWNTO 0
SIGNAL digit_flipper : STD_LOGIC_VECTOR(3 DOWNTO
;(0
BEGIN

Create a 1Hz 4-bit counter --
First, a large counter to produce a 1 second (approx) --
enable
from the 25 MHz Clock --
(PROCESS (CLOCK_50
BEGIN
IF ( [ ▼ ] )
THEN
```

سؤال 3

Correct

Mark 1.00 out of 1.00

:Given the the signals bellow

;SIGNAL s1: BIT
;(SIGNAL s2: BIT_VECTOR(7 DOWNTO 0
;SIGNAL s3: STD_LOGIC
;(SIGNAL s4: STD_LOGIC_VECTOR(7 DOWNTO 0
;(VARIABLE v1: BIT_VECTOR(7 DOWNTO 0

Select which statement bellow are legal (more than one (choice are allowed

:Select one or more

- [ ] ;;(s2 (5 DOWNTO 0);<= S2 (6 DOWNTO 2
- [ ] ;'s2 <= 'Z
- [x] ;"s2 <= "10101011 ✔
- [x] ;s2(3) <= s1 ✔

سؤال 4

Not answered

Marked out of 5.00

Use the following code as a component in your top level design to build a **3-bit Subtractor** and compile it in Quartus II and upload a print screen showing the successful compilation with the code. **The top level**

A. Design this circuit for 4 stages using VHDL **(10 Marks)**

B. The entity shown in dashed box has the interfaces as follows:

```
ENTITY MUXED_FF IS
PORT (clk, D, S, Load : IN STD_LOGIC;
      Q : OUT STD_LOGIC);
END ENTITY;
```

Assume you have this entity in your library. Write VHDL to design the same circuit assuming the number of stages is a **GENERIC** named **N** and using GENERATE statement. **(10 Marks)**

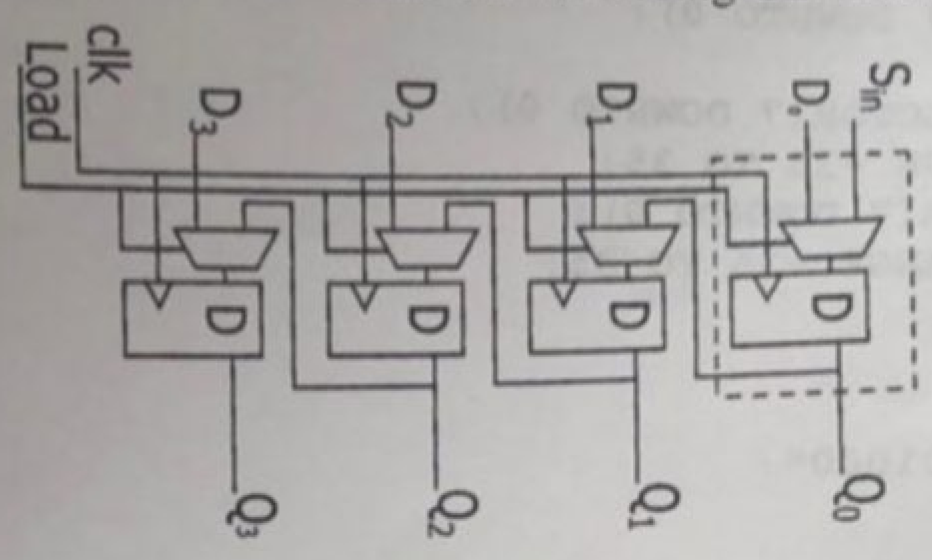


Figure Q3

Q1_Given the Testbench code bellow that aim to test a 8-bit shift register with serial input D, Clear input (SCLR operate on active high), and a clock (CLK). This shift register operate on +ev edge. Then what are the content of this register would be after 8 clock cycles with shift left:

```
LIBRARY ieee;
USE ieee.std_logic_1164.all;

ENTITY part4_vhd_tst IS
END part4_vhd_tst;
ARCHITECTURE part4_arch OF part4_vhd_tst IS
Clock_Cycle: time := 40ns;
SIGNAL Clk : STD_LOGIC;

SIGNAL SCLR : STD_LOGIC;
SIGNAL D : STD_LOGIC;

SIGNAL O : STD_LOGIC


        COMPONENT 4SRegister

                PORT (
                Clk D, SCLR: IN STD_LOGIC;
                O : OUT STD_LOGIC
                );

        END COMPONENT;


        BEGIN


        i1 : 4SRegister
        PORT MAP (
        -- list connections between master ports and signals
        Clk => Clk,
        D => D,
        SCLR => SCLR,
        O => O
        );


        init : PROCESS
        BEGIN

        SCLR<='0';
        D<='0';
        WAIT 10ns;
        D<='1';
        WAIT 15ns;
        D<='0';
        WAIT Clock_Cycle;
        D<='1';
```

```
init : PROCESS
BEGIN

SCLR<='0';
D<='0';
WAIT 10ns;
D<='1';
WAIT 15ns;
D<='0';
WAIT Clock_Cycle;
D<='1';

SCLR<='1';
WAIT Clock_Cycle;
D<='0';

WAIT Clock_Cycle;

D<='1';
WAIT 2*Clock_Cycle;

D<='0';

WAIT;
END PROCESS init;

Clock_1 : PROCESS

BEGIN

Clock_Loop: Loop
Clk<= '0';wait Clock_Cycle/2;

Clk<= '1';
wait Clock_Cycle/2;
end loop

END PROCESS;
END part4_arch;
```

Select one:

- a. 0000 0000
- b. 1011 0000
- c. 1011 0111
- d. 0011 0000
- e. 0000 1100

Given the signals bellow

;(signal X : std_logic_vector(7 downto 0

;(signal Y :unsigned(7 downto 0

Which of the following is a legal VHDL assignment
:Select one

;(a. X <=unsigned(Y

;(b. Y <=to_unsigned(201, 8

;(c. Y <=std_logic_vector(X

;(d. X <=to_integer(Y

سؤال 8
Incorrect
Mark 0.00 out of 1.00

Say that a(7:0) "00110011" and b(3:0) = "1111". Determine the values produced by the :assignments below

;a(7 DOWNTO 4) XOR NOT b

:Answer

0011 xor not 1111 ~~(0011) xor not (1111)~~

سؤال 9
Not answered
Marked out of 2.00

سؤال 5
Incorrect
Mark 0.00 out of 1.00

To design a 0 to 8 counter we will need how many D-flip flops?

Select one:

a. 3
b. 5
c. 4
d. 6

سؤال 6
Incorrect
Mark 0.00 out of 1.00

The following VHDL code is for:

```
entity XXX is
port (B, D, R : in STD_LOGIC;
Q: out STD_LOGIC);
end XXX;
architecture arch of XXX is
begin
X: process (B, R)
begin
```

```
;(
;END COMPONENT

BEGIN

i1 : 4SRegister
) PORT MAP
list connections between master ports and --
signals
,Clk => Clk
,D => D
,SCLR => SCLR
O => O
;(

init : PROCESS
BEGIN
;'SCLR<='0
;'D<='0
;WAIT 10ns
;'D<='1
;WAIT 15ns
;'D<='0
;WAIT Clock_Cycle
;'D<='1
;WAIT 2*Clock_Cycle
;'D<='0
;'SCLR<='1
;WAIT Clock_Cycle
;'D<='1
;WAIT 3*Clock_Cycle
```

; D<= 0
;WAIT Clock_Cycle

;WAIT
;END PROCESS init

Clock_1 : PROCESS

BEGIN

Clock_Loop: Loop
;'Clk<= '0
;wait Clock_Cycle/2
;'Clk<= '1
;wait Clock_Cycle/2
end loop

;END PROCESS
;END part4_arch
:Select one

a. 1011 0010

b. 1110 0001

c. 0000 0000

d. 1110 0000

e. 1011 0111

Question 1

Not yet answered

Marked out of 0.50

Flag question

Name the parts of the structure of a a VHDL file?

Select one:

- a. Library, Entity, and architecture
- b. Entity, and architecture
- c. Library, Entity, process, and architecture
- d. Library, Entity, Process, Components, and Architecture
- e. Entity

Question 2

Not yet answered

Marked out of 0.50

Flag question

Say that a(7 downto 0) "11110011" and b(3 downto 0) = "1001". Determine the values produced by the assignments below:

b(5 DOWNTO 2) XNOR (NOT b);

Answer:

Question 3

Not yet answered

Marked out of 0.50

Flag question

The initial value that is assigned to Generic in component could be overwritten in the top-level entity:

Select one:

- True
- False

Question 4

Not yet answered

Marked out of 3.00

Flag question

To test a 2x4 Decoder using Records, complete the following record declaration and its constant of array:

type test_vector is record

d : std_logic_vector( );

o : std_logic_vector ( );

end record;

... array ( ) of test_vector;

To test a 2x4 Decoder using Records, complete the following record declaration and its constant of array:

type test_vector is record

d : std_logic_vector(2 downto 0);

o : std_logic_vector (4 downto 0);

end record;

type test_vector_array is array (0 to 2) of test_vector;

constant test_vectors : test_vector_array := (

(, , , , ,),

(, , , , ,),

(, , , , ,),

(, , , , ,));

Determine which lines have syntax errors in the accompanying VHDL code:

(Note selecting wrong answers would cause a penalty)

```
1  library ieee;
2  use ieee.std_logic_1164.all;
3  entity Error_finder is
4     generic
5     (NUM_STAGES : natural := 16);
6     port
7     (  clk, enable, reset, sr_in:  std_logic;
8        sr_out     : out std_logic;);
9  end entity;
10 architecture rtl of Error_finder is
11    type sr_length is array ((NUM_STAGES-1) to 0) of std_logic;
12    signal sr: sr_length;
13 begin
14    process (clk, reset)
15    begin
16       if (reset = '1') then
17          sr <= (others=>'0');
18       elsif (rising_edge(clk)) then
19          if (enable = '1') then
20             sr((NUM_STAGES-1) downto 1) <= sr((NUM_STAGES-2) downto 0);
21             sr(0) <= sr_in;
22       end if;
23    end process;
24    sr_out <= sr(NUM_STAGES-1 downto 0);
25 end rtl;
```

Select one or more:

- a. Line 24
- b. Line 3
- c. line 8

- f. Line 11
- g. Line 4
- h. Line 18
- i. Line 6
- j. Line 13
- k. Line 16
- l. Line 15
- m. Line 2
- n. Line 10
- o. line 7
- p. Line 25
- q. Line 9

Question 2
Not yet answered
Marked out of 0.60
Flag question

To design a counter that counts the even numbers from 0 to 20, how many D-flip flops we will need ?

Select one:

- a. 5
- b. 6
- c. 3
- d. 4

Next page

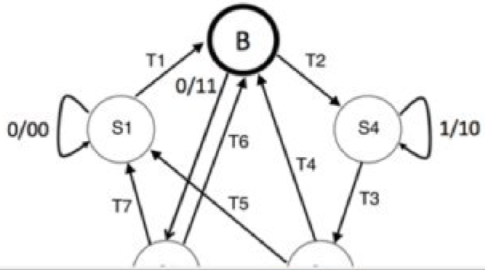
Q2_- **KEY0 is resetn, KEY1 is the clock for reg_A. Operation: set SW to the value desired for A, then store this value KEY1. A is shown on HEX3 and HEX2. Then change SW to the value desired for B, which is displayed on HEX1 and = A + B is displayed on HEX5 and HEX4, and the carry out is shown on LEDR0. Notice the the 7-segments leds turns given.**

```
LIBRARY ieee;
USE ieee.std_logic_1164.all;
USE ieee.std_logic_unsigned.all;


ENTITY part5 IS
PORT ( SW : IN STD_LOGIC_VECTOR(7 DOWNTO 0);
KEY : IN STD_LOGIC_VECTOR(1 DOWNTO 0);
LEDR : OUT STD_LOGIC_VECTOR(9 DOWNTO 0);
HEX5, HEX4, HEX3, HEX2, HEX1, HEX0 : OUT STD_LOGIC_VECTOR(0 TO 6));
END part5;

ARCHITECTURE Behavior OF part5 IS
COMPONENT hex7seg
PORT ( hex : IN STD_LOGIC_VECTOR(3 DOWNTO 0);
display : OUT STD_LOGIC_VECTOR(0 TO 6));
END COMPONENT;
COMPONENT regn
PORT ( R : IN STD_LOGIC_VECTOR(7 DOWNTO 0);
Clock, Resetn : IN STD_LOGIC;
Q : OUT STD_LOGIC_VECTOR(7 DOWNTO 0));
END COMPONENT;
SIGNAL A, B : STD_LOGIC_VECTOR(7 DOWNTO 0);
SIGNAL S : STD_LOGIC_VECTOR(8 DOWNTO 0);
SIGNAL carry : STD_LOGIC;
BEGIN
-- regn (R, Clock, Resetn, Q)
A_reg: regn PORT MAP (SW, KEY(1), KEY(0), A);
B <= SW;

-- drive the displays through 7-seg decoders
digit_3: hex7seg PORT MAP (A [(7 DOWNTO 0)], HEX3);
digit_2: hex7seg PORT MAP (A [(7 DOWNTO 0)], HEX2);
digit_1: hex7seg PORT MAP (B [ ], HEX1);
digit_0: hex7seg PORT MAP (B [ ], HEX0);


S <= [ ];
carry <= S(8);
digit_5: hex7seg PORT MAP (S [ ], HEX5);
digit_4: hex7seg PORT MAP (S [ ], HEX4);
LEDR(0) <= carry;
LEDR(9 DOWNTO 1) <= "000000000";
END Behavior;

LIBRARY ieee;
USE ieee.std_logic_1164.all;
ENTITY regn IS
PORT ( R : IN STD_LOGIC_VECTOR(7 DOWNTO 0);
Clock, Resetn : IN STD_LOGIC;
Q : OUT STD_LOGIC_VECTOR(7 DOWNTO 0));
END regn;

ARCHITECTURE Behavior OF regn IS
BEGIN
PROCESS (Clock, Resetn)
BEGIN
IF (Resetn = '0') THEN
Q <= (OTHERS => '0');
ELSIF (Clock'EVENT AND Clock = '1') THEN
Q <= R;
END IF;
END PROCESS;
END Behavior;

LIBRARY ieee;
USE ieee.std_logic_1164.all;

ENTITY hex7seg IS
PORT ( hex : IN STD_LOGIC_VECTOR(3 DOWNTO 0);
display : OUT STD_LOGIC_VECTOR(0 TO 6));
END hex7seg;

ARCHITECTURE Behavior OF hex7seg IS
BEGIN

PROCESS (hex)
BEGIN
CASE hex IS
WHEN "0000" => display <= "0000001";
WHEN "0001" => display <= "1001111";
WHEN "0010" => display <= "0010010";
WHEN "0011" => display <= "0000110";
WHEN "0100" => display <= "1001100";
WHEN "0101" => display <= "0100100";
WHEN "0110" => display <= ["0100000"];
WHEN "0111" => display <= "0001111";
WHEN "1000" => display <= "0000000";
WHEN "1001" => display <= "0000100";
WHEN "1010" => display <= "0001000";
WHEN "1011" => display <= "1100000";
WHEN "1100" => display <= "0110001";
WHEN "1101" => display <= "1000010";
WHEN "1110" => display <= "0110000";
WHEN OTHERS => display <= "0111000";
END CASE;
END PROCESS;
END Behavior;
```

Question 3
Not yet answered
Marked out of 1.50
Flag question

Using the state transition table as a guide, label each transition (T1-T7) with IN/OUT (ex: 0/00) to represent which input causes the given transition and what the output is.

S	IN	Next S	OUT
A	0	A	00
A	1	B	11
B	0	C	11
B	1	D	00
C	0	A	10
C	1	B	01
D	0	E	01
D	1	D	10
E	0	A	10
E	1	B	01



Quiz navigation
1 2 3 4 5 6 7 8
9 10 11 12 13 14 15 16
17 18 19 20 21 22 23 24
25
Finish attempt ...
Time left 2:49:54